

Flexible Deployment, Reliable Connectivity



ODU302 4G Outdoor Router

· 4G LTE

· Wi-Fi

· IP65 Rated

· Cloud Managed

1. Product Overview

The ODU302 is an all-in-one outdoor cellular router that combines field-proven reliability with a clean, modern design. Its smooth, streamlined enclosure fully houses all antennas, doing away with the clutter of external antennas for a neat, coordinated appearance that blends naturally into a wide range of outdoor commercial settings and open areas. The device integrates 4G cellular WAN, 2.4 GHz Wi-Fi LAN, dual wired Ethernet ports, multi-layer link redundancy, a full suite of encrypted VPNs, and unified cloud-based operations — delivering three core advantages: minimal on-site installation, reliable 24/7 operation in any weather, and large-scale batch deployment. Designed to address four key industry pain points — areas with no fixed broadband coverage, indoor cellular signal blockage, tight installation spaces, and the heavy burden of managing many distributed sites — the ODU302 is ideal for retail stores, self-service kiosks, warehouses and parks, outdoor surveillance, and small venues, providing a standardized all-in-one outdoor networking solution.

Core Features

- **4G High-Speed Access:** Supports 4G high-speed cellular access with built-in omnidirectional antennas for full, seamless signal coverage — greatly simplifying on-site deployment and ensuring a stable connection.
- **Wi-Fi:** Supports IEEE 802.11 b/g/n for reliable wireless networking.
- **Always Online:** Link backup, dual-SIM link switching, heartbeat monitoring, and a hardware watchdog keep communications uninterrupted.
- **Secure & Trustworthy:** Enterprise-grade firewall and a full suite of mainstream VPN tunnel protocols meet compliance requirements for encrypted commercial and transaction data.
- **Environmental Reliability:** IP65 all-weather protection and a wide operating range of -20 °C to +70 °C withstand scorching sun, extreme cold, dust, and rain.
- **Cloud Managed:** Connects to the Device Manager cloud platform for bulk remote management and operations.
- **Flexible Installation:** Compact and lightweight design supports wall, pole, and top mounting to suit every scenario.

- **Dual Redundant Power:** Supports both IEEE 802.3af PoE power over Ethernet and DC 9–36 V wide-voltage terminal power.

Typical Application Scenarios

Small Parks & Public Amenities

Small parks and outdoor public facilities typically have many scattered sites, where running wired cabling everywhere is prohibitively expensive and operations require centralized remote management. The ODU302 supports wall, pole, and top mounting to flexibly adapt to building walls and poles across various deployment sites. Its multi-layer link redundancy keeps services online 24/7. Compatible with the InHand cloud management platform, it enables centralized monitoring, batch configuration, and unified firmware pushes for up to thousands of terminals — combined with VPN tunnels to build a private remote network, it fully meets the needs of centralized operations for large-scale, distributed sites.

Warehouses & Logistics Parks

Steel-framed warehouse buildings tend to block cellular signals, limiting indoor router speeds, while on-site scanners and surveillance equipment need a mix of wired and wireless networking. The device mounts on exterior walls, rooftops, or park poles, connecting to the 4G network through built-in omnidirectional antennas and providing stable connectivity to on-site devices via Ethernet and Wi-Fi.

Street Retail Stores

Many storefronts along the street lack fiber access. Indoor 4G routers often suffer signal attenuation from wall blockage, and traditional external-antenna solutions disrupt the storefront's overall appearance. The ODU302 features a sleek, streamlined body that blends seamlessly into its surroundings and supports discreet wall or pole mounting outdoors with no protruding external structure. PoE delivers power and data over a single cable, simplifying on-site installation. Combining high-speed 4G cellular with 2.4 GHz Wi-Fi, it simultaneously carries POS systems, surveillance cameras, and guest Wi-Fi. Its web-based visual management interface is easy to use, so stores don't need dedicated IT staff for maintenance.

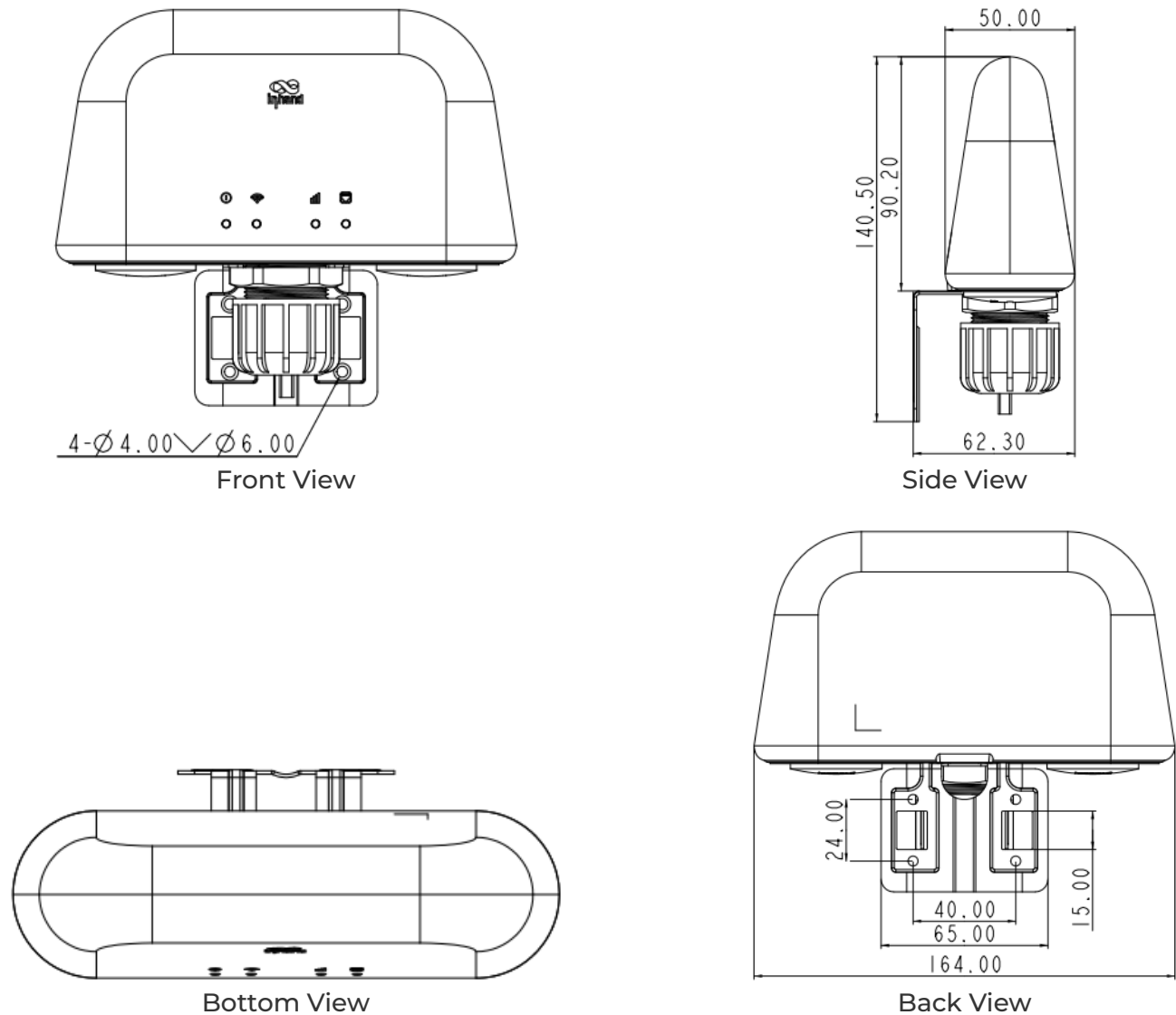
Outdoor Self-Service Kiosks

Outdoor terminals such as ATM kiosks and fuel dispensers have tight internal space. Traditional routers with external antennas take up valuable cabinet space and require additional cutouts and cabling. The ODU302 supports top mounting and is inherently IP65-rated for dust and water resistance, so it can be mounted directly on the outside of the equipment for a seamlessly integrated installation. Its built-in omnidirectional cellular antenna captures signals from all directions, ensuring stable, sustained network quality.

Outdoor Security Surveillance

Outdoor surveillance points are typically deployed on exposed poles, where wired broadband is usually unavailable. Traditional approaches place the router in a separate waterproof enclosure, which can shield cellular signals and degrade network quality. The ODU302 has native IP65 dust and water resistance and a wide operating range of -20 °C to +70 °C, so no additional enclosure is needed — it can be mounted outdoors directly. Outdoor deployment with built-in omnidirectional antennas yields stronger cellular signals, keeping surveillance video streams uploaded reliably.

Product Dimensions



Notes:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. Dimensions (L × W × H): 164 × 50 × 129.7 mm (6.46 × 1.97 × 5.11 in).
3. All dimensions are approximate and **for reference only**.
4. Drawings **must not be used for manufacturing**.
5. Specifications may change without prior notice.



2. Product Highlights

Highlight 1: Reliability Meets Aesthetics

The ODU302 is engineered for reliable all-weather outdoor operation, wrapped in a sleek, fully sealed, streamlined design with all antennas built in to eliminate the clutter of external accessories. Its soft, minimalist body blends naturally into storefronts, self-service kiosks, and park environments — striking a balance between hard-core networking capability and visual fit for each setting.

Highlight 2: Lightweight Multi-Mode Installation for Every Scenario

The compact, lightweight body supports three mounting options — wall, pole, and top — covering storefront exteriors, park poles, and outdoor self-service equipment. For ATM and similar kiosks, a dedicated top-mount solution attaches the unit to the outside of the equipment, taking up none of the cabinet's tight interior space, reducing cabinet cutouts and internal cabling, and keeping the terminal's exterior clean and consistent without disturbing the site's appearance.

Highlight 3: Enterprise-Grade Security & Cloud Operations at Scale

An enterprise-grade stateful inspection firewall and a full suite of mainstream VPN tunnel protocols meet compliance requirements for encrypted commercial transactions and remote private network data. Backed by the InHand Device Manager cloud platform, hundreds to thousands of distributed devices can be managed centrally — remote configuration, firmware upgrades, traffic statistics, and fault alerts — dramatically cutting manual on-site maintenance costs across multiple locations.

Highlight 4: Multi-Layer Link Redundancy for Uninterrupted Uptime

Three layers of protection work together: hardware watchdog self-recovery, dual-SIM cellular backup, and cellular/wired dual-link redundancy. Weak SIM signals, dial failures, or wired broadband outages all trigger seamless failover to the backup channel, keeping business running without interruption — ideal for transaction terminals, warehouse equipment, and outdoor surveillance that demand high uptime.

Highlight 5: Dual Redundant Power, No Single Point of Failure

The device supports both IEEE 802.3af PoE and DC 9–36 V wide-voltage terminal power, with the two inputs monitored in parallel and millisecond-level automatic switching if either fails. It adapts to storefront network-cable power, equipment low-voltage cabinets, and solar power setups — perfectly suited to unattended, always-on deployments.

3. Hardware Specifications

3.1 Hardware Overview

Parameter	Specification
Processor	580 MHz
Memory	128 MB DDR2 SDRAM
Storage	32 MB SPI NOR Flash
4G Module	4G LTE Cat.4, DL 150 Mbps / UL 50 Mbps
Recommended Users	32

Parameter	Specification
Ethernet Ports	2 × 10/100 Mbps RJ45, WAN/LAN switchable. LAN2 supports PoE
SIM Slots	2 × Nano SIM (drawer-style), dual SIM failover
Digital I/O	2 × configurable DI or DO
4G Antennas	2 × built-in 4G antennas (ANT_4G_1, ANT_4G_2)
Wi-Fi Antennas	1 × built-in Wi-Fi antenna
Wi-Fi	IEEE 802.11 b/g/n (Wi-Fi 4), 2.4 GHz, up to 150 Mbps
Wi-Fi TX Power	802.11b: 16 dBm ± 2 dBm (11 Mbps) 802.11g: 16 dBm ± 2 dBm (54 Mbps) 802.11n@2.4 GHz HT20 MCS7: 16 dBm ± 2 dBm 802.11n@2.4 GHz HT40 MCS7: 16 dBm ± 2 dBm
LED Indicators	System, Wi-Fi, Cellular, Ethernet
Reset Button	Pinhole reset button
Power Supply	DC 9–36 V (4-pin 2×2 combined power & I/O terminal, over-current / reverse-polarity protected) IEEE 802.3af PoE input
Standby Power	90~110 mA @ 12 V
Operating Power	280~300 mA @ 12 V
Peak Power	340~360 mA @ 12 V
Protection Rating	IP65
Enclosure Material	Plastic

Parameter	Specification
Operating Temperature	-20 °C ~ +70 °C (-4 °F ~ +158 °F)
Storage Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Relative Humidity	5% ~ 95%, non-condensing
Dimensions	164 × 50 × 129.7 mm (6.46 × 1.97 × 5.11 in)
Weight	Approx. 500 g (1.10 lb)
Mounting	Wall, pole, top-mount

4. Network Connectivity

4.1 Cellular Network

- **Network Standards:** GSM/GPRS/EDGE, UMTS/HSPA+/EVDO, TD-SCDMA, FDD LTE/TDD LTE
- **Network Access:** APN, VPDN
- **Authentication:** CHAP/PAP
- **Connection Modes:** Always-on, on-demand, manual dial
- **Activation:** Data trigger, SMS trigger

4.2 SIM Management

- **Dual-SIM Support:** Drawer-style dual Nano SIM card slots
- **Dual SIM Failover:** Primary/backup SIM auto-switch support
- **Smart Switching:**
 - Automatic SIM switch on dial failure
 - Automatic switch when signal falls below threshold
 - Configurable primary/backup card and max dial attempts
- **SIM Binding:** ICCID binding, PIN code management

4.3 Wired Network

- **WAN Protocol:** Static IP, DHCP, PPPoE
- **LAN Protocol:** ARP, Ethernet
- **Interface Mode:** 2 × RJ45 ports, WAN/LAN configurable
- **NAT:** Network address translation (NAPT)

4.4 Wi-Fi 4 Wireless Access

- **Standard:** IEEE 802.11 b/g/n (Wi-Fi 4)
- **Frequency:** 2.4 GHz single-band
- **Max Data Rate:** 150 Mbps
- **Modes:** Access Point (AP), Client (STA)
- **Security:** Open, Shared Key, WPA/WPA2
- **Encryption:** WEP/TKIP/AES

5. Link Management & Backup

5.1 Intelligent Link Backup

- **Backup Modes:** Cellular dial-up and WAN port can serve as primary/backup for each other
- **Probe Methods:** ICMP/TCP/DNS link quality probing (configurable)
- **Switching Policies:** Automatic failover on link failure, auto-redial on disconnection

5.2 VRRP Hot Backup

- **Protocol Support:** VRRP (Virtual Router Redundancy Protocol)
- **Use Case:** Dual-device hot standby for improved business continuity

5.3 Link Quality Assurance

- **Link Detection:** Heartbeat packet monitoring, auto-reconnect on disconnection

- **Dual SIM Switching:** Automatic backup SIM switch on cellular link or dial-up anomaly
- **Built-in Watchdog:** Device self-check with fault auto-recovery

6. Security Features

6.1 Firewall

- **Stateful Inspection:** Stateful packet inspection (SPI)
- **DoS Protection:** Denial-of-service attack prevention
- **Access Control:** Access Control List (ACL)
- **Content Filtering:** URL-based content filtering
- **Port Mapping:** Port forwarding, virtual IP mapping
- **IP-MAC Binding:** IP-MAC binding; multicast/Ping packet filtering
- **NAT:** NAT support

6.2 VPN Support

- IPsec VPN (IKEv1, IKEv2)
- L2TP VPN (client)
- PPTP VPN (client)
- GRE Tunnel
- OpenVPN (client/server)
- WireGuard
- ZeroTier
- CA digital certificate support

6.3 Traffic Shaping

- **Bandwidth Control:** Interface-level bandwidth limiting
- **IP Rate Limiting:** Per-IP upload/download rate limiting
- **QoS Policy:** Bandwidth and rate limit policies

7. Local Network Services

7.1 DHCP Service

- **DHCP Server:** Dynamic IP address assignment
- **DHCP Relay:** Cross-subnet DHCP relay
- **DHCP Client:** Client address acquisition

7.2 DNS Service

- **DNS Relay/Proxy:** Local caching and forwarding
- **Custom DNS:** Configurable DNS server

7.3 Dynamic DNS (DDNS)

- **Multi-Provider Support:** Major DDNS service providers
- **Automatic Updates:** Automatic domain resolution update on WAN/cellular IP change

7.4 Static Routing

- **Route Entries:** Static route configuration
- **Default Routes:** Cellular/WAN default route support

7.5 IP Passthrough

- **Bridge Mode:** WAN/dial-up IP pass-through to downstream device
- **Flexible Configuration:** Passthrough mode on dial-up and WAN ports

8. Cloud Management & Operations

The ODU302 is powered by the InHand Device Manager cloud management platform, enabling unified remote operations for large-scale distributed outdoor devices.

8.1 Device Manager Cloud Management

Feature	Description
Device Onboarding	Device Manager cloud platform and custom server support
Remote Management	Device status query, remote configuration push, firmware upgrade, remote reboot
Batch Operations	Mass device configuration and maintenance
SNMP	SNMP v1/v2c/v3 with Trap support
Traffic Management	Traffic thresholds, statistics, and alarms
Alert Types	System reboot, LAN port up/down, traffic alarm, SIM card fault
Notification	Email alerts (configurable)

9. System Management & Maintenance

9.1 Device Management

- **Web Management:** Graphical web configuration interface
- **CLI Management:** Telnet, SSH, Console

- **Configuration Import/Export:** Backup and restore
- **Factory Reset:** Pinhole reset or web factory reset
- **Firmware Upgrade:** Web or Device Manager remote upgrade

9.2 Logs & Diagnostics

- **System Logs:** Local syslog, remote log, serial log export; persistent critical logs on power loss
- **Diagnostic Tools:** Ping (network connectivity), Traceroute (route tracing), Network speed test (link performance)

9.3 SMS Functions

- **Status Query:** SMS query device status
- **Remote Reboot:** SMS-triggered device reboot

9.4 Other System Capabilities

- **Dial-on-Demand:** On-demand dial; data/SMS activation
- **Status Query:** System status, modem status, network connection status, routing status
- **Scheduled Maintenance:** NTP time synchronization; auto-recovery on fault (built-in watchdog)

10. Ordering Information

10.1 Product Models

Model	Region	Description
ODU302-CNC4	China	LTE-FDD: B1/B3/B5/B8 LTE-TDD: B34/B38/B39/B40/B41 WCDMA: B1/B8 GSM: B3/B8 CDMA: BC0 TD-SCDMA: B34/B39
ODU302-NAC4	North America	LTE-FDD: B2/B4/B5/B12/B13/B14/B66/B71 WCDMA: B2/B4/B5
ODU302-EUC4	Europe/APAC	LTE-FDD: B1/B3/B5/B7/B8/B20/B28 LTE-TDD: B38/B40/B41 WCDMA: B1/B5/B8 GSM/EDGE: B3/B8

10.2 Package Components

- ODU302 outdoor unit × 1
- IEEE 802.3af PoE power adapter × 1
- 1 m RJ45 cable × 1
- Mounting kit (wall/pole/top-mount) × 1
- Quick Start Guide × 1

11. Reliability Standards & Certifications

11.1 Protection Rating

- IP Rating: IP65 (dust-tight, protected against high-pressure water jets)

11.2 Mechanical Reliability

Test Item	Standard
Shock	IEC 60068-2-27
Vibration	IEC 60068-2-6
Free Fall	IEC 60068-2-32

11.3 EMC Immunity

Test Item	Standard
ESD	EN 61000-4-2, Level 2
Radiated RF	EN 61000-4-3, Level 2
EFT/Burst	EN 61000-4-4, Level 2
Surge	EN 61000-4-5, Level 2
Conducted RF	EN 61000-4-6, Level 2
Oscillatory Wave	EN 61000-4-12, Level 2
Power Frequency Magnetic	EN 61000-4-8, 400 A/m

11.4 Certifications & Compliance

- FCC
- IC
- CE
- PTCRB
- AT&T
- Verizon
- T-Mobile

Note: Certification status is current as of publication. Check with InHand sales for the latest updates.

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